

2018 Online Mathematics exam (simulated)

General Instructions: This simulated exam contains 60 multiple-choice questions. Choose the most correct answer for each question. All work should be strictly your own.

1. Consider Matrix $A = \begin{pmatrix} -1 & 2 & 1 \\ 3 & 4 & 2 \\ 1 & 1 & 5 \end{pmatrix}$. Which of the following statements is true about the cofactor C_{ij} and minor M_{ij} of the entries of A?

- A. $C_{12} = M_{12}$
- B. $C_{23} = -M_{23}$
- C. $M_{11} = -10$
- D. $C_{22} = -M_{22}$

2. What is the transpose of the matrix $M = \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & 3 \\ 2 & 1 & 5 \end{pmatrix}$?

- A. $\begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 1 \\ 1 & 3 & 5 \end{pmatrix}$
- B. $\begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 5 \end{pmatrix}$
- C. $\begin{pmatrix} 2 & 1 & 5 \\ 0 & -1 & 3 \\ 1 & 2 & 1 \end{pmatrix}$
- D. $\begin{pmatrix} 1 & 2 & 2 \\ 0 & -1 & 1 \\ 1 & 3 & 5 \end{pmatrix}$

3. Which one of the following sets is the domain of the rational function $f(x) = \frac{x^3+x^2}{x^4+2x^2-3}$?

- A. $\mathbb{R} \setminus \{1\}$
- B. $\mathbb{R} \setminus \{-1, 1\}$
- C. $\mathbb{R} \setminus \{-3, 1\}$
- D. $\mathbb{R} \setminus \{-1, 1, -3\}$

4. Which one of the following statements correctly defines a circle?
- A. The set of points in a plane whose distances from two fixed points sum to a constant.
 - B. The set of points in a plane equidistant from a fixed point and a fixed line.
 - C. The set of points in a plane that are at a fixed distance from a given point.
 - D. The set of points in a plane whose difference of distances from two fixed points is constant.
5. Which of the following statements is true in borrowing money?
- A. The borrower only pays back the principal amount.
 - B. The lender assumes no risk when giving out a loan.
 - C. Borrowing money involves an agreement to repay the principal amount along with accrued interest.
 - D. Collateral is always required for any type of financial borrowing.
6. What is the correct form for the partial fraction decomposition of the rational expression $\frac{3x^2+5x-1}{(x+2)(x+1)^2}$?
- A. $\frac{A}{x+2} + \frac{B}{(x+1)^2}$
 - B. $\frac{A}{x+2} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$
 - C. $\frac{A}{x+2} + \frac{Bx+C}{(x+1)^2}$
 - D. $\frac{A}{x+2} + \frac{B}{x+1} + \frac{Cx+D}{(x+1)^2}$
7. Let $f(x) = \frac{2x+1}{x-1}$. Which of the following equations is the equation of the tangent line to the graph of f at $(2, f(2))$?
- A. $3x + y = 11$
 - B. $y - 3x = -1$
 - C. $x + 3y = 17$
 - D. $3x - y = 1$
8. What is the measure of each interior angle of a regular polygon with 15 sides?
- A. 144°
 - B. 150°
 - C. 156°
 - D. 162°

9. Let $f(x) = x^3 + 4x$ and $g(x) = 3x^3 - x^2$. What is the derivative of $(f + g)(x)$?
- A. $12x^2 - 2x + 4$
 - B. $4x^3 - x^2 + 4x$
 - C. $12x^2 - x + 4$
 - D. $9x^2 - 2x + 4$
10. Consider the quadrilateral ABCD where $A = (0, 0)$, $B = (-1, 1)$, $C = (0, 2)$, and $D = (1, 1)$. Which one of the following statements is true about this quadrilateral?
- A. It is a rectangle but not a square.
 - B. It is a square.
 - C. It is a trapezoid.
 - D. Its adjacent sides are unequal in length.
11. Which of the following is considered a primary source of a cash loan for a business?
- A. Accumulated depreciation
 - B. Retained earnings
 - C. Commercial Banks
 - D. Accounts receivable
12. Let $A = \{x \in \mathbb{R} : x^3 - x = 0\}$ and $B = \{x \in \mathbb{Z} : -1 \leq x < 3\}$. What is the relative complement of B with respect to A (i.e., $A \setminus B$)?
- A. $\{2\}$
 - B. $\{0, 1\}$
 - C. $\emptyset$
 - D. $\{-1, 0, 1\}$
13. Given the set $B = \{\emptyset, \{-2, 0\}\}$, what is the power set of B ?
- A. $\{\emptyset, \{-2, 0\}\}$
 - B. $\{\emptyset, \{\emptyset\}, \{\{-2, 0\}\}, \{\emptyset, \{-2, 0\}\}\}$
 - C. $\{\{\emptyset\}, \{\{-2, 0\}\}\}$
 - D. $\{\emptyset, \{-2\}, \{0\}, \{-2, 0\}\}$

14. What is the simplified form of $\log_{16}(4) - \log_2(\sqrt{12.5}) + \log_{0.5}(2)$?
- A. $\log_2(5)$
 - B. $1 - \log_2(5)$
 - C. $-\log_2(5)$
 - D. 0
15. Which statement is true about the graphs of the exponential functions $f(x) = a^x$ and $g(x) = \left(\frac{1}{a}\right)^x$, where $a > 1$?
- A. They are reflections of each other across the x-axis.
 - B. They are reflections of each other across the y-axis.
 - C. They never intersect each other.
 - D. Both functions are monotonically increasing.
16. Which of the following statements is NOT true about the mean of a given data set?
- A. It is highly affected by extreme values (outliers).
 - B. The sum of the deviations of the items from the mean is always zero.
 - C. It must always be one of the actual data values present in the set.
 - D. It uses every value in the data set for its calculation.
17. Which one of the following is NOT a power function?
- A. $f(x) = x^2$
 - B. $f(x) = x^{1/3}$
 - C. $f(x) = x^{-2}$
 - D. $f(x) = 3^x$
18. Which one of the following numbers is an irrational number?
- A. 3.14159 (terminating)
 - B. $\sqrt{16}$
 - C. $\frac{22}{7}$
 - D. π

19. What is the area of the parallelogram spanned by the vectors $\vec{u} = i - 2j + k$ and $\vec{v} = 2i + 2j - k$?
- A. $\sqrt{14}$
 - B. $3\sqrt{5}$
 - C. 9
 - D. $\sqrt{35}$
20. What is the fundamental period of the function $f(x) = \sin(x)$?
- A. $\pi/2$
 - B. π
 - C. 2π
 - D. 4π
21. What is the simplified form of the expression $\left(\sqrt{45} + \frac{1}{\sqrt{5}}\right)(\sqrt{5} - \sqrt{20})$?
- A. 16
 - B. -16
 - C. 8
 - D. -8
22. In statistical sampling, if a population is divided into homogeneous groups and a random sample is drawn from each group, this method is known as:
- A. Simple random sampling
 - B. Cluster sampling
 - C. Stratified sampling
 - D. Systematic sampling
23. A mathematics exam contains 10 multiple-choice questions and 3 short-answer questions. According to the multiplication principle, how many ways can a student select exactly one multiple-choice question and exactly one short-answer question to solve first?
- A. 13
 - B. 30
 - C. 10^3
 - D. $\binom{13}{2}$

24. Using the Binomial Theorem, what is the coefficient of the term containing x^3y^4 in the expansion of $(x - 2y)^7$?
- A. 35
 - B. -560
 - C. 560
 - D. -280
25. A translation vector maps the origin $(0, 0)$ to the point $(-2, 4)$. Under this same translation, what is the image of the point $(3, 1)$?
- A. $(5, -3)$
 - B. $(1, 5)$
 - C. $(-6, 4)$
 - D. $(-5, 3)$
26. If the initial point of a vector is $A(2, -1, 4)$ and the terminal point is $B(4, 3, 2)$, what is the position vector $\vec{AB}$?
- A. $\langle -2, -4, 2 \rangle$
 - B. $\langle 6, 2, 6 \rangle$
 - C. $\langle 2, 4, -2 \rangle$
 - D. $\langle 8, -3, 8 \rangle$
27. Let vector $\vec{u}$ have an initial point at $(1, 2)$ and a terminal point at $(-3, 6)$. What are the components of the vector $-\frac{1}{2}\vec{u}$?
- A. $\langle 2, -2 \rangle$
 - B. $\langle -2, 2 \rangle$
 - C. $\langle 4, -4 \rangle$
 - D. $\langle 1, -1 \rangle$
28. Consider the rational function $f(x) = \frac{x^2-4}{x-2}$. The domain of f is the set of all real numbers EXCEPT:
- A. -2
 - B. 2
 - C. 0
 - D. The domain is all real numbers.

29. Which of the following properties of matrices and scalar multiplication r is TRUE for any two matrices A and B of the same dimension?
- A. $r(AB) = (rA)(rB)$
 - B. $r(A + B) = rA + rB$
 - C. $(A + B)^T = A^T - B^T$
 - D. $\det(A + B) = \det(A) + \det(B)$
30. If A is a 3×3 square matrix, which of the following is always true about its transpose A^T ?
- A. $\det(A^T) = -\det(A)$
 - B. $\det(A^T) = \frac{1}{\det(A)}$
 - C. $\det(A^T) = \det(A)$
 - D. $A^T = A^{-1}$
31. Let A be a 3×3 invertible matrix with determinant $|A| = 4$. What is the determinant of $2(A^{-1})$?
- A. $\frac{1}{2}$
 - B. 2
 - C. 8
 - D. $\frac{1}{4}$
32. In the real number system, there is a fundamental property stating there is a one-to-one correspondence between the set of real numbers and:
- A. The set of rational numbers.
 - B. The set of points on a continuous straight line.
 - C. The set of natural numbers.
 - D. The set of prime numbers.
33. What is the solution set for the radical equation $\sqrt{x^2 + 3x} = 2$?
- A. $\{1, -4\}$
 - B. $\{1, 4\}$
 - C. $\{-1, 4\}$
 - D. $\{-1, -4\}$

34. Using logarithm rules, which expression is equivalent to $\log_3(9x^2)$?
- A. $2\log_3(3x)$
 - B. $2 + 2\log_3(x)$
 - C. $9\log_3(x^2)$
 - D. $3 + 2\log_3(x)$
35. Given the exponential functions $f(x) = 2^x$ and $g(x) = 0.5^x$, which statement describes their monotonicity?
- A. Both functions are increasing.
 - B. Both functions are decreasing.
 - C. $f(x)$ is decreasing and $g(x)$ is increasing.
 - D. $f(x)$ is increasing and $g(x)$ is decreasing.
36. For the greatest integer (floor) function $f(x) = \lfloor x - 1 \rfloor$, which statement is true regarding its domain and range?
- A. Domain is $\mathbb{Z}$, Range is $\mathbb{R}$.
 - B. Domain is $\mathbb{R}$, Range is $\mathbb{Z}$.
 - C. Domain is $\mathbb{R}$, Range is all positive integers.
 - D. Domain is $\mathbb{Z}$, Range is $\mathbb{Z}$.
37. In an arithmetic progression, the first term is 11 and the common difference is -13 . What is the sum of the first 7 terms?
- A. -28
 - B. -196
 - C. -67
 - D. 196
38. In a geometric sequence, the first term is $g_1 = 2$ and the fifth term is $g_5 = 8$ (assume the common ratio is positive). What is the value of the 13th term (g_{13})?
- A. 32
 - B. 64
 - C. 128
 - D. 256

39. The infinite repeating decimal $0.373737\ldots$ can be expressed as which rational fraction in lowest terms?
- A. $\frac{37}{100}$
 - B. $\frac{37}{99}$
 - C. $\frac{37}{90}$
 - D. $\frac{1}{3}$
40. According to the properties of integration, which of the following is equivalent to $\int(3f(x) - g(x))dx$?
- A. $3 \int f(x)dx \cdot \int g(x)dx$
 - B. $\int 3f(x)dx - \int g(x)dx$
 - C. $3(\int f(x)dx - \int g(x)dx)$
 - D. $\int f(x)dx - 3 \int g(x)dx$
41. Three vertices of a parallelogram ABCD are $A(0, 0)$, $B(5, 0)$, and $C(7, 4)$. If AD is parallel to BC, what are the coordinates of vertex D ?
- A. $(2, 4)$
 - B. $(-2, 4)$
 - C. $(12, 4)$
 - D. $(2, -4)$
42. If α and β are complementary angles ($\alpha + \beta = 90^\circ$), which of the following trigonometric relations is always true?
- A. $\sin(\alpha) = \sin(\beta)$
 - B. $\tan(\alpha) = \tan(\beta)$
 - C. $\sin(\alpha) = \cos(\beta)$
 - D. $\sec(\alpha) = \cos(\beta)$
43. For the ungrouped data set: 12, 15, 12, 18, 20, 15, 12, 22, what is the mode?
- A. 15
 - B. 18
 - C. 12
 - D. 20

14. In descriptive statistics, the third decile (D_3) corresponds to which percentile?
- A. 3rd percentile
 - B. 30th percentile
 - C. 33rd percentile
 - D. 70th percentile
15. For a dataset with a maximum value of 60 and a minimum value of 20, what is the coefficient of range?
- A. 40
 - B. 0.5
 - C. 2.0
 - D. 0.33
16. Which formula defines the coefficient of mean deviation along the median?
- A. $\frac{\text{Mean Deviation}}{\text{Mean}}$
 - B. $\frac{\text{Mean Deviation}}{\text{Median}}$
 - C. $\frac{\text{Median}}{\text{Mean Deviation}}$
 - D. $\frac{\text{Standard Deviation}}{\text{Median}}$
17. If every value in a dataset is multiplied by a constant $k = 4$, the new mean will be:
- A. The original mean plus 4
 - B. Unchanged
 - C. Four times the original mean
 - D. The original mean divided by 4
18. A mathematical relation is strictly defined as a function if:
- A. Every element in the range maps to exactly one element in the domain.
 - B. Every element in the domain maps to exactly one element in the range.
 - C. It passes the horizontal line test.
 - D. The domain and range are equal sets.

49. Which one of the following intervals is the solution set of the inequality $|2x - 3| < 7$?

- A. $(-2, 5)$
- B. $(-5, 2)$
- C. $(-2, 7)$
- D. $(-7, 3)$

50. If p and q are propositions with truth values True and False respectively, which of the following is True?

- A. $p \wedge q$
- B. $p \implies q$
- C. $q \implies p$
- D. $\sim p \vee q$

51. What is the value of the sum $\sum_{k=1}^4 (2k - 1)$?

- A. 15
- B. 16
- C. 20
- D. 12

52. Which of the following defines the inverse of the function $f(x) = \frac{x}{x-1}$?

- A. $f^{-1}(x) = \frac{x-1}{x}$
- B. $f^{-1}(x) = \frac{x}{x-1}$
- C. $f^{-1}(x) = \frac{-x}{x-1}$
- D. $f^{-1}(x) = \frac{1}{x-1}$

53. The vertex of the parabola given by the equation $y = 2x^2 - 8x + 5$ is:

- A. $(2, -3)$
- B. $(-2, 21)$
- C. $(2, 5)$
- D. $(-2, -3)$

54. Which of the following sets is NOT closed under addition?

- A. The set of even integers
- B. The set of rational numbers
- C. The set of odd integers
- D. The set of real numbers

55. Let $z = 3 - 4i$. What is the modulus of the complex number z ?

- A. -1
- B. 5
- C. 7
- D. 25

56. If $\lim_{x \rightarrow 2} f(x) = 5$ and $\lim_{x \rightarrow 2} g(x) = -3$, what is $\lim_{x \rightarrow 2} [2f(x) - g(x)]$?

- A. 7
- B. 13
- C. 4
- D. -1

57. The derivative of $y = e^{3x} \sin(x)$ with respect to x is:

- A. $e^{3x} \cos(x)$
- B. $3e^{3x} \sin(x) + e^{3x} \cos(x)$
- C. $3e^{3x} \cos(x)$
- D. $e^{3x}(3 \sin(x) - \cos(x))$

58. Evaluate the definite integral $\int_0^1 x^2 \sqrt{x^3 + 1} \, dx$.

- A. $\frac{2}{9}(2\sqrt{2} - 1)$
- B. $\frac{2}{3}(2\sqrt{2} - 1)$
- C. $\frac{1}{3}$
- D. $\frac{2\sqrt{2}}{9}$

59. The unit vector in the direction of $\vec{v} = -3\mathbf{i} + 4\mathbf{j}$ is:

- A. $\frac{-3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$
- B. $-3\mathbf{i} + 4\mathbf{j}$
- C. $\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}$
- D. $\frac{-3}{25}\mathbf{i} + \frac{4}{25}\mathbf{j}$

50. If two dice are rolled simultaneously, what is the probability that the sum of the numbers is 8?

- A. $\frac{1}{6}$
- B. $\frac{5}{36}$
- C. $\frac{1}{9}$
- D. $\frac{7}{36}$

ANSWER KEY & EXPLANATIONS

1. **B** - The cofactor is defined as $C_{ij} = (-1)^{i+j}M_{ij}$. For $i = 2, j = 3, i + j = 5$ (odd), so $C_{23} = -M_{23}$.
2. **B** - The transpose of a matrix swaps its rows and columns. Row 1 becomes Column 1, etc.
3. **B** - The denominator factors as $(x^2 + 3)(x - 1)(x + 1)$. The real roots are 1 and -1. So domain is $\mathbb{R} \setminus \{-1, 1\}$.
4. **C** - A circle is geometrically defined as the set of all points in a plane equidistant from a single fixed center point.
5. **C** - Borrowing inherently implies the obligation to repay the borrowed principal amount plus the agreed-upon interest.
6. **B** - Since the denominator has a linear factor $(x + 2)$ and a repeated linear factor $(x + 1)^2$, the standard decomposition form is $A/(x + 2) + B/(x + 1) + C/(x + 1)^2$.
7. **A** - $f'(x) = \frac{-3}{(x-1)^2}$. At $x = 2, f'(2) = -3$. The point is $(2, 5)$. The line is $y - 5 = -3(x - 2) \implies 3x + y = 11$.
8. **C** - Interior angle formula is $\frac{180(n-2)}{n}$. For $n = 15$, it is $\frac{180 \times 13}{15} = 156^\circ$.
9. **A** - $(f + g)(x) = 4x^3 - x^2 + 4x$. Its derivative is $12x^2 - 2x + 4$.
10. **B** - Checking the distance between all adjacent points gives $\sqrt{2}$, and adjacent slopes are negative reciprocals (perpendicular), making it a square.
11. **C** - Commercial banks are primary institutional sources for direct cash loans.
12. **C** - $A = \{-1, 0, 1\}$. $B = \{-1, 0, 1, 2\}$. The relative complement of B with respect to A refers to elements in A that are not in B ($A \setminus B$), which is empty ($\emptyset$).
13. **B** - A set with 2 elements has $2^2 = 4$ subsets: the empty set, the whole set, and singletons of its elements.
14. **C** - $\log_{16}(4) = 0.5, \log_{0.5}(2) = -1, \log_2(\sqrt{12.5}) = \log_2(5) - 0.5$. Evaluating $0.5 - (\log_2(5) - 0.5) - 1 = -\log_2(5)$.
15. **B** - $f(x) = a^x$ and $g(x) = a^{-x}$. Substituting $-x$ for x creates a reflection across the y-axis.
16. **C** - The mean is calculated from the data but does not necessarily have to be one of the actual data point values.
17. **D** - $f(x) = 3^x$ is an exponential function, not a power function (which must be of the form x^p).
18. **D** - π cannot be written as a simple fraction, making it an irrational number. $22/7$ is a rational approximation.
19. **B** - The area is the magnitude of the cross product $\vec{u} \times \vec{v}$. $|\langle 0, 3, 6 \rangle| = \sqrt{9 + 36} = \sqrt{45} = 3\sqrt{5}$.
20. **C** - The standard sine function completes one full cycle every 2π radians.
21. **B** - Expand: $(\frac{16\sqrt{5}}{5}) \times (-\sqrt{5}) = -16$.
22. **C** - Dividing a population into homogeneous groups (strata) and randomly selecting from each is stratified sampling.
23. **B** - By the fundamental counting principle, 10 choices $\times$ 3 choices = 30 combinations.
24. **C** - The term is $\binom{7}{4}x^3(-2y)^4 = 35x^3(16y^4) = 560x^3y^4$.

25. **B** - The translation vector is $\vec{T} = \langle -2 - 0, 4 - 0 \rangle = \langle -2, 4 \rangle$. Image of $(3, 1)$ is $(3 - 2, 1 + 4) = (1, 5)$.
26. **C** - Position vector $\vec{AB} = B - A = \langle 4 - 2, 3 - (-1), 2 - 4 \rangle = \langle 2, 4, -2 \rangle$.
27. **A** - $\vec{u} = \langle -3 - 1, 6 - 2 \rangle = \langle -4, 4 \rangle$. Multiplying by $-1/2$ gives $\langle 2, -2 \rangle$.
28. **B** - The rational function is undefined when the denominator is 0, which occurs at $x = 2$.
29. **B** - Scalar multiplication distributes over matrix addition.
30. **C** - The determinant of a matrix and the determinant of its transpose are always equal.
31. **B** - For an $n \times n$ matrix, $|kA| = k^n|A|$. Thus, $|2A^{-1}| = 2^3|A^{-1}| = 8 \times (1/4) = 2$.
32. **B** - The Completeness Axiom guarantees every point on a continuous number line corresponds exactly to one real number.
33. **A** - Squaring both sides: $x^2 + 3x = 4 \implies x^2 + 3x - 4 = 0 \implies (x - 1)(x + 4) = 0$.
34. **B** - $\log_3(9x^2) = \log_3(9) + \log_3(x^2) = 2 + 2\log_3(x)$.
35. **D** - Base $2 > 1$ (increasing), whereas base $0.5 < 1$ (decreasing).
36. **B** - You can input any real number, but the floor function always outputs an integer.
37. **B** - $S_n = \frac{n}{2}(2a + (n - 1)d)$. For $n = 7$: $3.5(22 + 6(-13)) = 3.5(22 - 78) = 3.5(-56) = -196$.
38. **C** - $g_5 = g_1 \times r^4 \implies 8 = 2r^4 \implies r^2 = 2$. $g_{13} = g_1 \times r^{12} = 2 \times (2)^3 = 2 \times 64 = 128$.
39. **B** - Let $x = 0.3737\dots$, then $100x = 37.3737\dots$. Subtracting yields $99x = 37 \implies x = 37/99$.
40. **B** - Integrals are linear operators; constants pull out and sums/differences split.
41. **A** - In a parallelogram, $\vec{AD} = \vec{BC}$. $\vec{BC} = \langle 7 - 5, 4 - 0 \rangle = \langle 2, 4 \rangle$. So, $D = A + \vec{BC} = (2, 4)$.
42. **C** - Co-functions of complementary angles are equal (sine and cosine).
43. **C** - 12 appears three times, which is more frequent than any other value.
44. **B** - Deciles divide data into tenths. D_3 is exactly the 30th percentile.
45. **B** - Coefficient of Range = $\frac{\text{Max} - \text{Min}}{\text{Max} + \text{Min}} = \frac{60 - 20}{60 + 20} = \frac{40}{80} = 0.5$.
46. **B** - It is defined as the mean deviation relative to the central measure (in this case, median).
47. **C** - Multiplying every observation in a dataset by a constant multiplies the mean by that same constant.
48. **B** - The definition of a function guarantees one unique output (range) for every valid input (domain).
49. **A** - $-7 < 2x - 3 < 7 \implies -4 < 2x < 10 \implies -2 < x < 5$.
50. **C** - A conditional statement $q \implies p$ is True if the premise q is False.
51. **B** - $(2(1) - 1) + (2(2) - 1) + (2(3) - 1) + (2(4) - 1) = 1 + 3 + 5 + 7 = 16$.
52. **B** - $y = \frac{x}{x-1} \implies yx - y = x \implies x(y - 1) = y \implies x = \frac{y}{y-1}$.
53. **A** - $x = \frac{-b}{2a} = \frac{8}{4} = 2$. $y = 2(4) - 16 + 5 = -3$.
54. **C** - Odd + Odd = Even (which is outside the set of odd integers).
55. **B** - Modulus = $\sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5$.
56. **B** - $2(5) - (-3) = 10 + 3 = 13$.
57. **B** - Apply product rule: $(e^{3x})' \sin(x) + e^{3x}(\sin(x))' = 3e^{3x} \sin(x) + e^{3x} \cos(x)$.
58. **A** - Substitute $u = x^3 + 1$, $du = 3x^2 dx$. Integration leads to $\frac{2}{9}(x^3 + 1)^{3/2}$ from 0 to 1.
59. **A** - Magnitude is $\sqrt{(-3)^2 + 4^2} = 5$. Unit vector is $\frac{\vec{v}}{5}$.
60. **B** - Ways to get 8: (2,6), (3,5), (4,4), (5,3), (6,2). Total 5 ways out of 36.